

LAB 3

ARCHITECTURAL JUSTIFICATION

Problem Statement #09 — FACULTY RESEARCH GRANT & PUBLICATION TRACKER

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1. Objective

Evaluate architectural styles, select the most appropriate architecture for the Faculty Research Grant & Publication Tracker, and document the architectural decision.

2. Architectural Style Analysis

Layered Architecture: separates the system into Presentation, Business, and Data layers. It provides clear separation of concerns and is suitable for a structured portal application.

Microservices Architecture: provides independently deployable services and independent scaling, but introduces operational complexity and data-consistency challenges.

Client-Server Architecture: provides centralized control and simple deployment, but can create a scalability bottleneck and a single point of failure.

3. Architecture Selection

We chose **Layered Architecture** for the Faculty Research Grant & Publication Tracker.

4. Components Identified

- **Web Portal UI** – Faculty Researcher screens and Research Dean dashboard.
- **Authentication & Access** – authentication and role-based access.
- **Grant & Expense Manager** – grant records, expenses, and expense-cap validation.
- **Publication Manager** – publications, indexing status, and citations.
- **Approval Workflow** – co-author approval and approval-status tracking.
- **Analytics Service** – fund burn-up and research/publication performance metrics.
- **Database Repository** – grants, expenses, publications, and users.
- **Audit Ledger** – immutable, tamper-evident audit records.

5. Interfaces

- **IGrantService** – grant management.
- **IPublicationService** – publication management.
- **IApprovalReview** – approval review.
- **IAnalyticsService** – analytics and reporting.
- **IApprovalRequest** – internal approval request.
- **IDataAccess** – parameterised database access.
- **IAuditLog** – append-only audit logging.

6. Written Justification

Reason 1 – Separation of Concerns: Grant management, expense validation, publication management, approval workflow, and analytics are separated into focused business components, making the system easier to maintain.

Reason 2 – Clear Component Interaction: The portal communicates with business services through defined interfaces, while business components access the database and audit ledger through dedicated interfaces.

Security Advantage: The immutable Audit Ledger provides a tamper-evident record of financial approvals and publication-status modifications. Authentication and role-based access protect system operations.

Performance Benefit: Separating presentation, business processing, and data access keeps responsibilities organized and supports responsive access during expected peak usage.

7. Requirements Covered

- **FR-01:** Grant records and approved funding limits.
- **FR-02:** Research expenses, balance calculation and expense-limit validation.
- **FR-03:** Publication and citation/indexing records.
- **FR-04:** Co-author approval workflow.
- **FR-05:** Research Dean summaries of grant utilization, pending approvals and publication performance.
- **NFR-01:** Immutable/tamper-evident audit trail.
- **NFR-02:** Responsive access under expected peak usage.